

Is Your Effect Real?

PS50008A Week 14 | Lecture

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Learning Outcomes

By the end of this lecture you will be able to:

- Explain the logic of null hypothesis significance testing
- Interpret p-values correctly (and avoid common misconceptions)
- Understand confidence intervals as ranges of plausible values
- Calculate and interpret Cohen's d
- Choose and conduct independent vs paired t-tests
- Report complete results in APA format

Opening

Return to Week 12

Remember the DataLab Results?

Lab A (abstract words): $M = 5.4$ words recalled

Lab B (concrete words): $M = 6.8$ words recalled

Difference: 1.4 words

The questions we need to answer:

1. Is this difference REAL, or just chance?
2. How CERTAIN are we about these estimates?
3. How BIG is this effect in practical terms?

The Three Questions

Today we learn to answer:

1. **Is it real?** → Hypothesis testing & p-values
2. **How certain?** → Confidence intervals
3. **How big?** → Effect sizes (Cohen's d)

You need ALL THREE for your Research Report.

Act 1: The Logic of Testing

NHST (~25 min)

The Fundamental Question

When we observe a difference in our data...

Is it a real effect, or just random chance?

The 1.4 word difference could reflect:

- A genuine effect of word type on memory
- OR just random variation in who was in each lab

Null Hypothesis Significance Testing (NHST)

NHST: A statistical framework that tests whether observed data are consistent with a null hypothesis of “no effect.”

We assume there's nothing going on, then ask: “How surprising is our data?”

The Two Hypotheses

Null Hypothesis (H_0)

- Assumes no effect/difference
- The “skeptic’s position”
- What we test against

“Word type has no effect on memory.”

Alternative Hypothesis (H_1)

- Assumes there IS an effect
- The researcher’s prediction
- What we hope to support

“Word type does affect memory.”

Summary

Null Hypothesis (H0)

$$\mu_1 = \mu_2$$

The Skeptic's View: There is no real difference. Any gap you see in the data is just random noise/luck. The drugs are equally effective.

Alternative Hypothesis (H1)

$$\mu_1 \neq \mu_2$$

The Believer's View: The means are actually different. The difference is systematic and represents a real effect in the population.

The Logic: Proof by Contradiction

1. **Assume H_0 is true** (no real effect)
2. **Calculate** how likely our data would be under H_0
3. **If our data would be very unlikely...**
4. **Reject H_0** and accept H_1

Like a court trial: Assume innocence (H_0) until evidence (data) proves guilt beyond reasonable doubt.

Act 2: The p-value

What It Really Means (~20 min)

What is a p-value?

p-value: The probability of obtaining results at least as extreme as those observed, **assuming the null hypothesis is true.**

It answers: “If there were truly no effect, how often would we see data this extreme just by chance?”

Example: Memory Data

If our memory comparison has $p = 0.03$, this means:

“If word type truly had no effect on memory, we would observe a difference this large (or larger) in only **3% of experiments**, just by chance.”

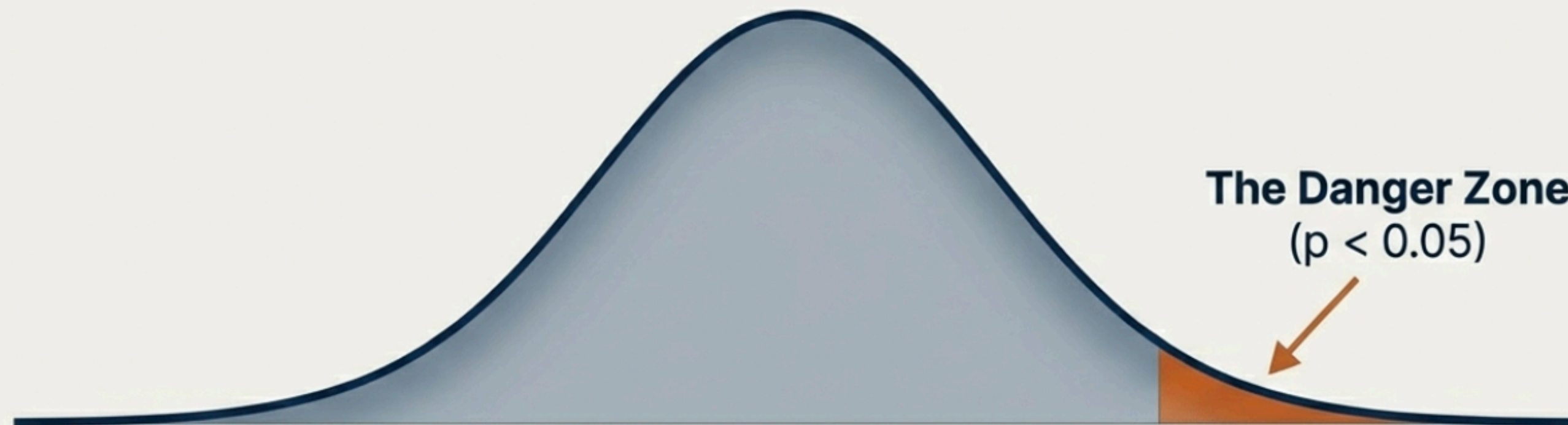
Because 3% is rare, we conclude H_0 is probably wrong.

The Significance Threshold (α)

Alpha (α): The predetermined threshold for rejecting H_0 . Conventionally set at **0.05 (5%)**.

p-value	Decision
$p < .05$	Reject H_0 (“significant”)
$p \geq .05$	Fail to reject H_0 (“not significant”)

A visual demonstration



$p > 0.05$ (Fail to reject Null)

The result is common. It could easily happen by luck. We assume no real difference exists.

$p < 0.05$ (Reject Null)

The result is rare. It is statistically significant. We conclude a real systematic difference exists.

The p-value is not a proof of truth, but a filter for luck.

What p-values Do NOT Mean



Common Misconceptions

The p-value is **NOT**:

- The probability that H_0 is true
- The probability the results occurred by chance
- The probability of replication
- A measure of effect size

A p-value only tells you how surprising your data would be IF H_0 were true.

Type I and Type II Errors

	H₀ True	H₀ False
Reject H₀	Type I Error (false positive)	Correct!
Fail to reject H₀	Correct!	Type II Error (false negative)

- **Type I error rate** = α (usually .05)
- **Type II error rate** = β (related to power)

BREAK

10 minutes

Act 3: How Big? How Certain?

Effect Sizes & Confidence Intervals (~25 min)

Beyond “Is There an Effect?”

The Limitation of p-values

A p-value tells us IF an effect probably exists.

It does NOT tell us:

- How BIG the effect is
- How PRECISE our estimate is

We need additional tools: **effect sizes** and **confidence intervals**.

Effect Sizes: How Big?

Cohen's d: A standardised measure of the difference between two means, expressed in standard deviation units.

$$d = \frac{\bar{X}_1 - \bar{X}_2}{S_{pooled}}$$

Key insight: Effect sizes are independent of sample size. A tiny effect can be “significant” with huge n.

Cohen's Benchmarks

d value	Interpretation	Visual Overlap
0.2	Small	85% overlap
0.5	Medium	67% overlap
0.8	Large	53% overlap

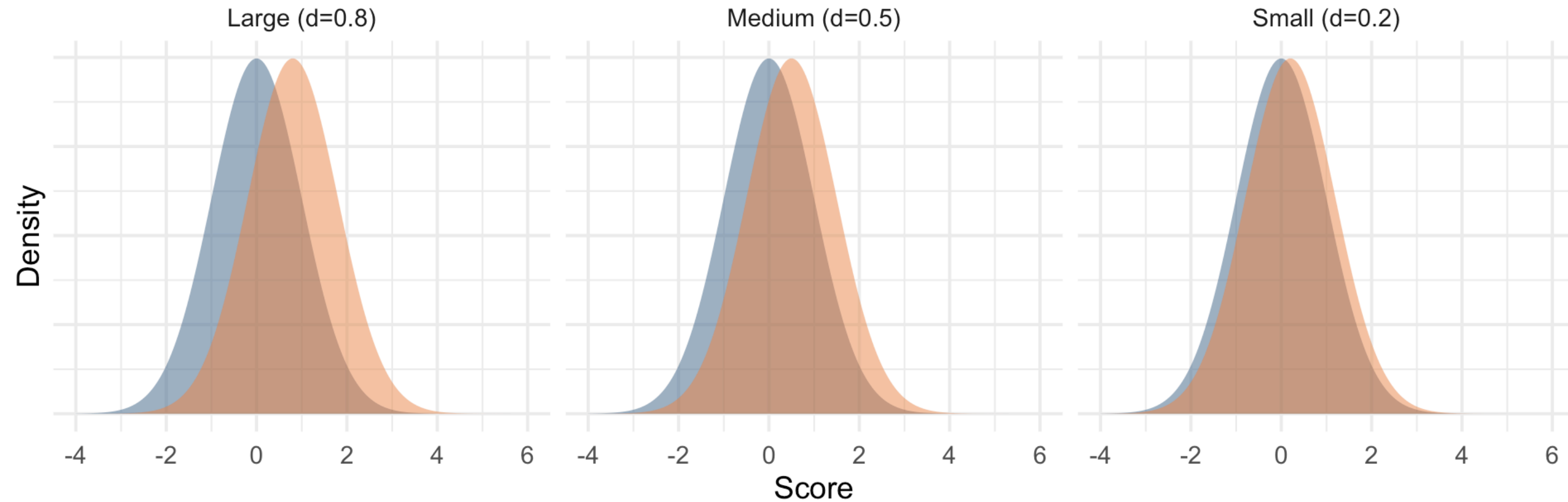
Note

These are rough guidelines. Context matters! In some fields, $d = 0.3$ is impressive.

Visualising Effect Sizes

What Different Effect Sizes Look Like

Blue = Control group, Orange = Treatment group



DataLab Example: Cohen's d

Our memory data:

- Lab A: $M = 5.4$, $SD = 1.8$
- Lab B: $M = 6.8$, $SD = 2.0$

$$d = \frac{6.8 - 5.4}{1.9} = \frac{1.4}{1.9} = 0.74$$

This is a **medium-to-large effect**.

Confidence Intervals: How Certain?

Confidence Interval (CI): A range of values that is likely to contain the true population parameter.

Instead of saying “the mean is 6.8”, we say:

“The mean is 6.8, **95% CI [6.07, 7.53]**”

This shows our **precision**.

Interpreting 95% CIs

Common Misconception

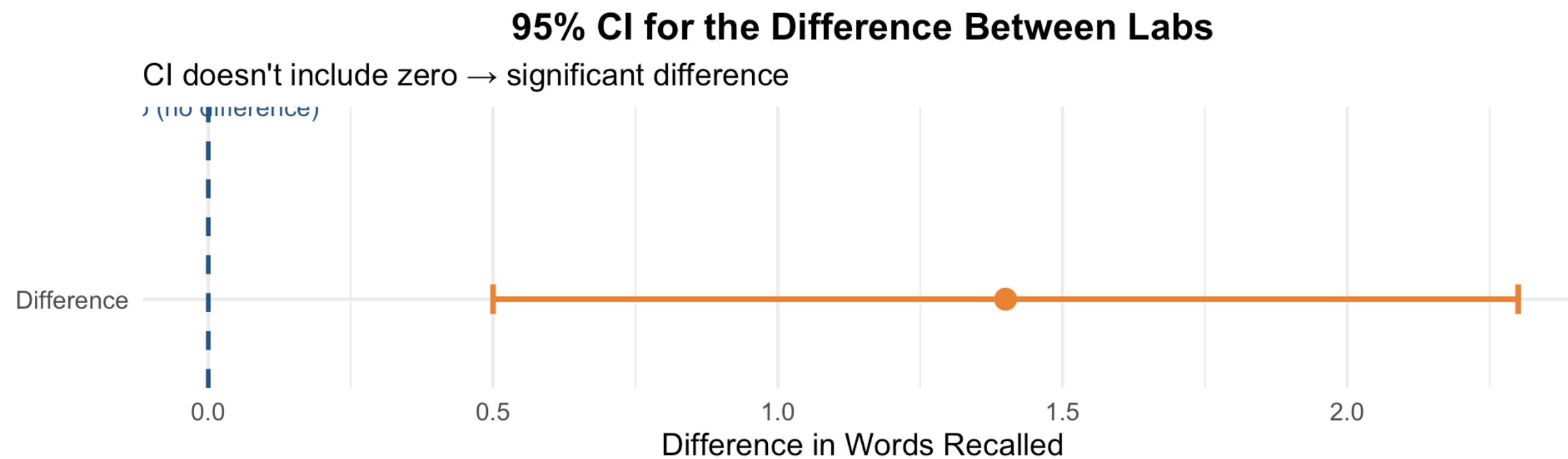
A 95% CI does NOT mean “there’s a 95% probability the true value is in this interval.”

Correct Interpretation

If we repeated this study many times and calculated a 95% CI each time, **95% of those intervals would contain the true value.**

CIs and Significance

The CI rule: If the 95% CI for a difference **does not include zero**, the difference is statistically significant at $p < .05$.



The Complete Picture

Report all three pieces:

1. **p-value:** Is the effect real?
2. **Effect size (d):** How big is it?
3. **95% CI:** How precise is our estimate?

Example: “The difference was significant, $t(58) = 2.86$, $p = .006$, $d = 0.74$, 95% CI [0.42, 2.38]”

Act 4: The t-tests

Independent vs Paired (~25 min)

Two Types of t-test

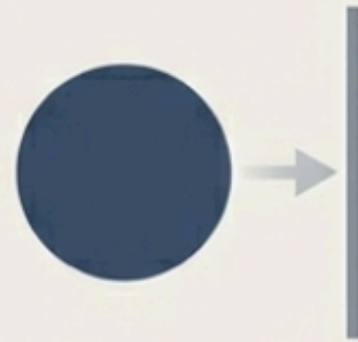
Independent samples t-test: Compares means from **two different groups**

Paired samples t-test: Compares means from the **same people** measured twice

The choice depends on your **design** (Week 13!).

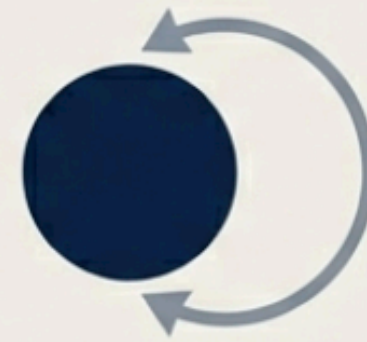
One or Two Tailed

One-Sample t-Test



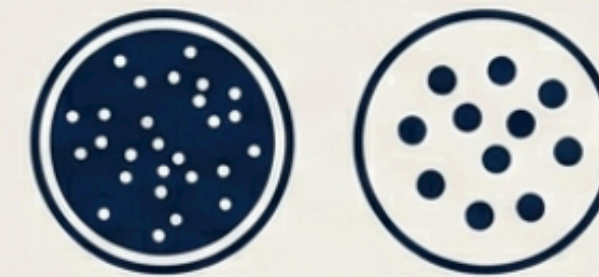
Compares one group against a standard.

Paired Samples t-Test



Compares the same group at different times (Before vs. After).

Independent Two-Sample t-Test



Compares the means of two entirely separate groups (e.g., Treatment vs. Control).

Independent t-test: When?

Use when you have:

- Two separate groups of participants
- Each participant in only ONE condition
- Between-subjects design

DataLab example: Lab A (abstract) vs Lab B (concrete)

Different students in each lab → Independent t-test

Independent t-test: The Formula

$$t = \frac{\bar{X}_1 - \bar{X}_2}{SE_{difference}}$$

The t-statistic is the ratio of:

- **Signal:** The difference between group means
- **Noise:** The standard error of that difference

Larger t = more evidence against H_0

Paired t-test: When?

Use when you have:

- The same participants in both conditions
- Each participant measured twice
- Within-subjects design

DataLab example: Pre-session mood vs Post-session mood

Same students measured at both times → Paired t-test

Paired t-test: The Formula

$$t = \frac{\bar{D}}{SE_{\bar{D}}}$$

Where $\bar{D}$ is the **mean of the difference scores**.

We're testing: Is the average change within each person significantly different from zero?

Why Paired Tests Are More Powerful

- **Controls for individual differences**
- Each person serves as their own control
- Reduces “noise” in the data
- Can detect smaller effects with same sample size



Tip

This is why Week 13's design decision matters! Within-subjects = more power.

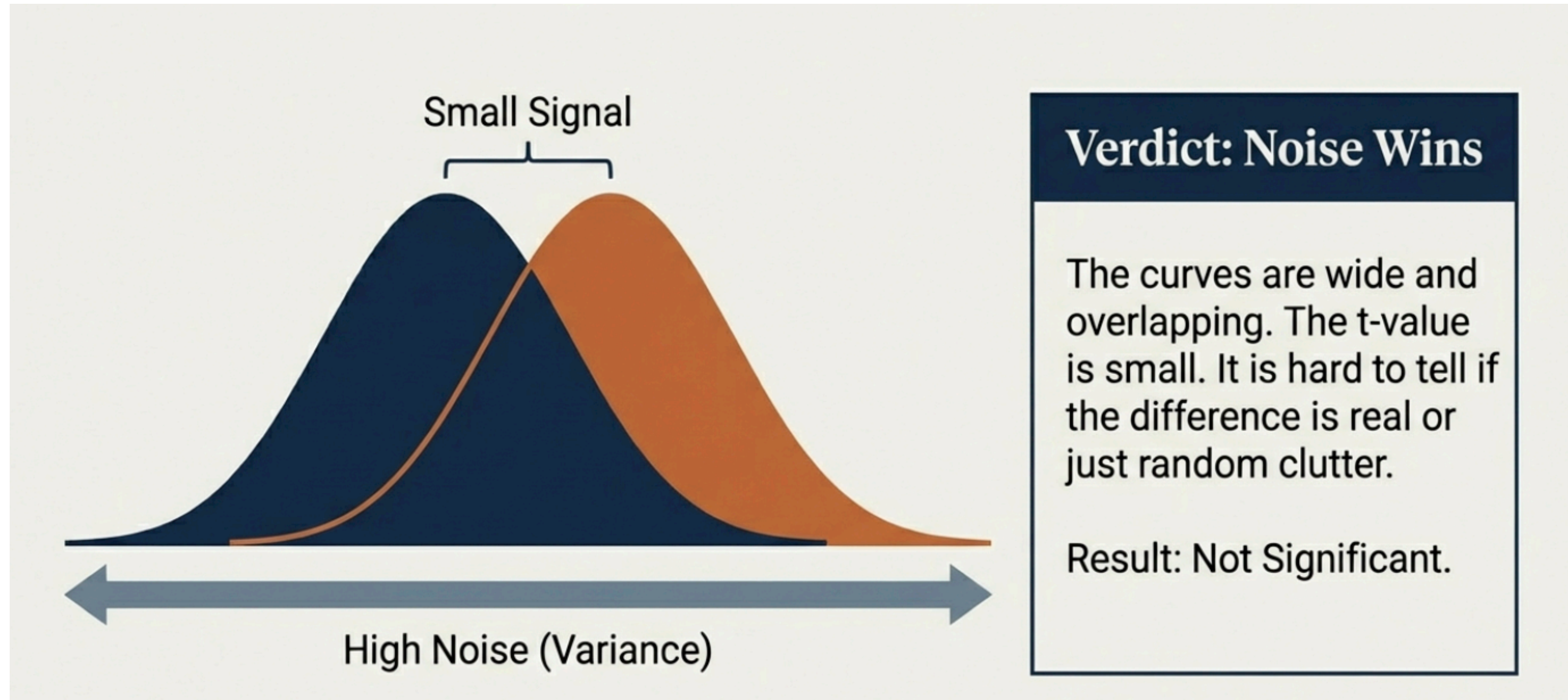
The Design → Test Link (Recap)

Your Design	Data Structure	Your Test
Between-subjects	Two independent groups	Independent t-test
Within-subjects	Paired scores	Paired t-test

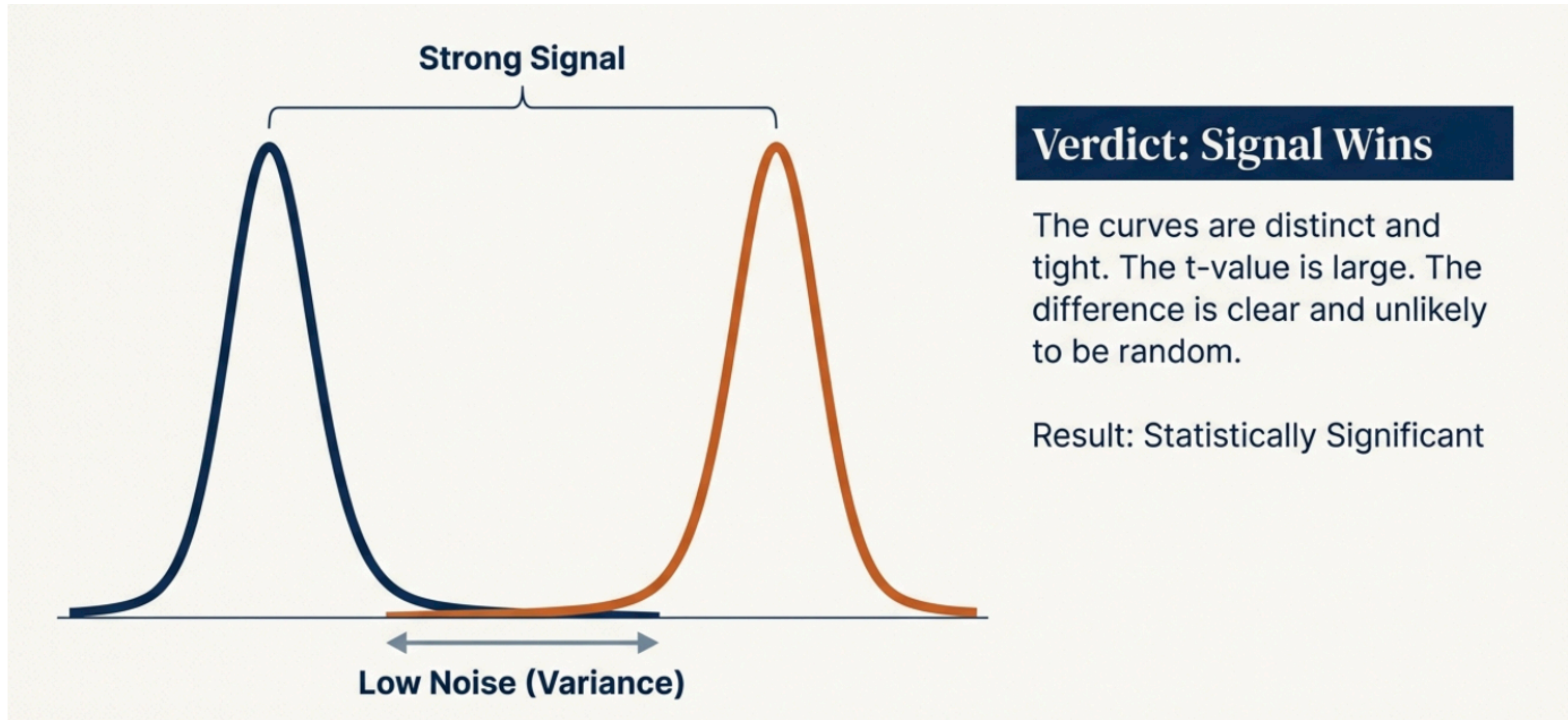
Warning

Using the wrong test gives invalid results!

In short



Signal wins



Closing: The Complete Picture

APA Reporting (~15 min)

What to Report

APA format for t-tests:

$t(df) = X.XX, p = .XXX, d = X.XX, 95\% \text{ CI } [X.XX, X.XX]$

All five pieces:

1. Test statistic (t)
2. Degrees of freedom (df)
3. p-value (exact, unless $p < .001$)
4. Effect size (Cohen's d)
5. Confidence interval

Example: Independent t-test

“An independent samples t-test revealed that participants who studied concrete words ($M = 6.8$, $SD = 2.0$) recalled significantly more words than those who studied abstract words ($M = 5.4$, $SD = 1.8$), $t(58) = 2.86$, $p = .006$, $d = 0.74$, 95% CI [0.42, 2.38].”

Example: Paired t-test

“A paired samples t-test showed that participants’ mood ratings were significantly higher after the DataLab session ($M = 6.5$, $SD = 1.4$) compared to before ($M = 5.3$, $SD = 1.6$), $t(29) = 3.64$, $p = .001$, $d = 0.67$, 95% CI [0.54, 1.86].”

Another view

Direction of test Type of test

A two-tailed t-test for independent samples showed that the difference between the Drug A group and Drug B group was not statistically significant, $t(58) = -2.16$, $p = .056$.

t-value and degrees of freedom Probability
(Non-significant because $> .05$)

In future

You will discuss assumptions...

-  **1. Suitable Structure:** Two completely separate, unrelated groups.
-  **2. Continuous Scale:** The variable must be measurable (e.g., pain score, time, weight).
-  **3. Normality:** Data should follow the approximate bell curve shape.
-  **4. Homogeneity of Variance:** The 'spread' in both groups should be roughly equal.
-  **5. Random Sampling:** Data must be a fair representation, not distorted by selection bias.

Connection to Your Assessments

Research Report Requirements

Your Results section must include:

- Descriptive statistics (M, SD) for each group/condition
- The appropriate t-test (independent or paired)
- Complete APA reporting: t, df, p, d, and 95% CI
- Interpretation in words

Key Takeaways

- **NHST** tests whether data are consistent with “no effect”
- **p-values** indicate probability of data given H_0 is true
- **Effect sizes (d)** tell us HOW BIG the effect is
- **Confidence intervals** tell us HOW CERTAIN we are
- **Independent t-tests** for between-subjects designs
- **Paired t-tests** for within-subjects designs
- **Report all three:** p-value + effect size + CI

Next Week

Week 15: Correlation

- Relationships between continuous variables
- Scatterplots and Pearson's r
- Correlation $\neq$ causation
- From correlation to regression

Questions?

Questions?

See you in the lab!